

Daily Budget vs. Daily Spending

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Project Overview



Analyze personal spending patterns

Compare planned vs. actual finances



Two-week data collection period

Daily tracking of budget and expenses



Personal finance relevance

Improve financial awareness and decision-making

Research Question & Hypotheses

Research Question

Is there a significant difference between planned budget and actual spending?

Null Hypothesis (H_0)

$$H_0 : \mu_d = 0$$

No difference between budget and spending

Alternative Hypothesis (H_1)

$$H_a : \mu_d \neq 0$$

Significant difference exists



	A	B	C
1	Date	Daily_Budget	Daily_Spending
2	4/6/2025	42	64.4
3	4/7/2025	39	61.2
4	4/8/2025	43	35.35
5	4/9/2025	47	44.8
6	4/10/2025	38	28.47
7	4/11/2025	38	33.54
8	4/12/2025	47	56.16
9	4/13/2025	43	63.25
10	4/14/2025	37	110.12
11	4/15/2025	42	29.74
12	4/16/2025	37	30.87
13	4/17/2025	37	3.45
14	4/18/2025	41	30.15
15	4/19/2025	30	34

Data Collection Methodology

Duration & Frequency

14 consecutive days

Daily recording, same time each evening

Variables Measured

Daily budget (USD): planned spending

Daily actual spending (USD): receipts and transactions

Collection Tools

Excel spreadsheet for recording

Mobile banking app for verification

Daily Comparison: Line Plot



Visualization Choice

Line plot shows temporal patterns



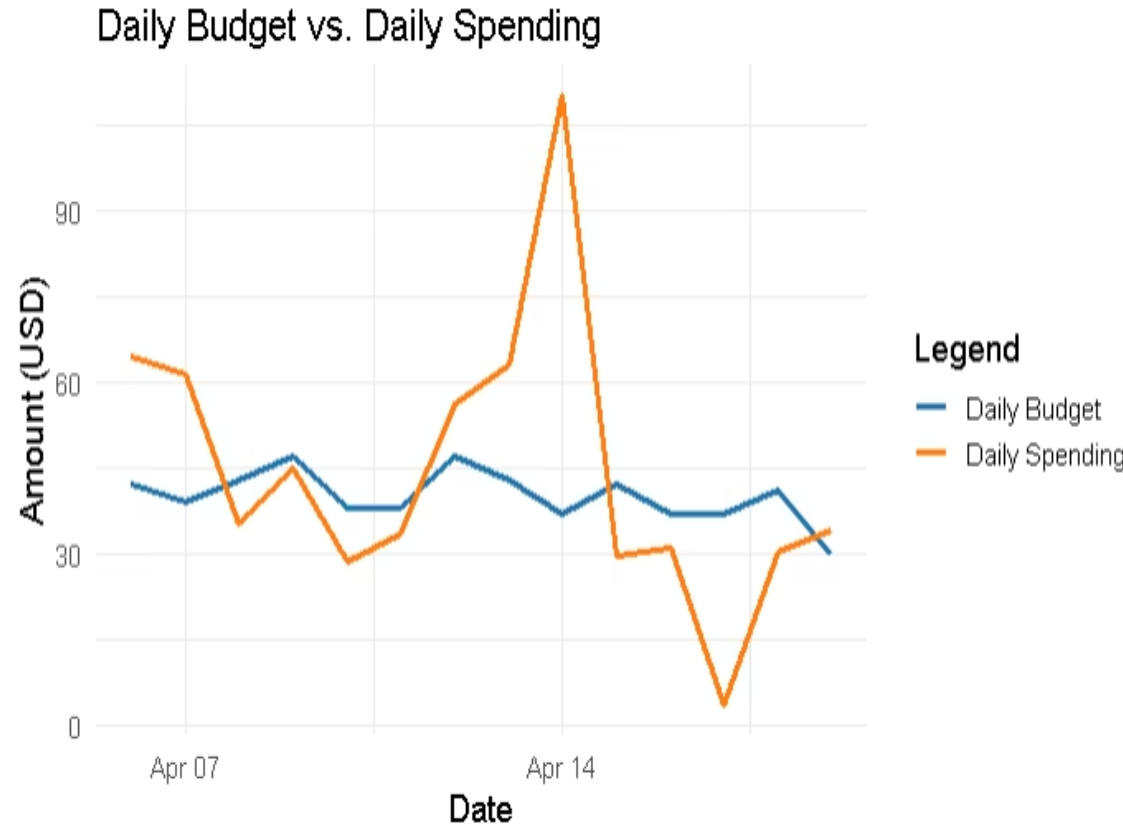
Key Observations

Spending exceeds budget on weekends



Patterns Revealed

Spending more volatile than budget



Daily Comparison: Bar Chart

Visual Encoding

X-axis = Days (1-14)

Y-axis = Amount (USD)

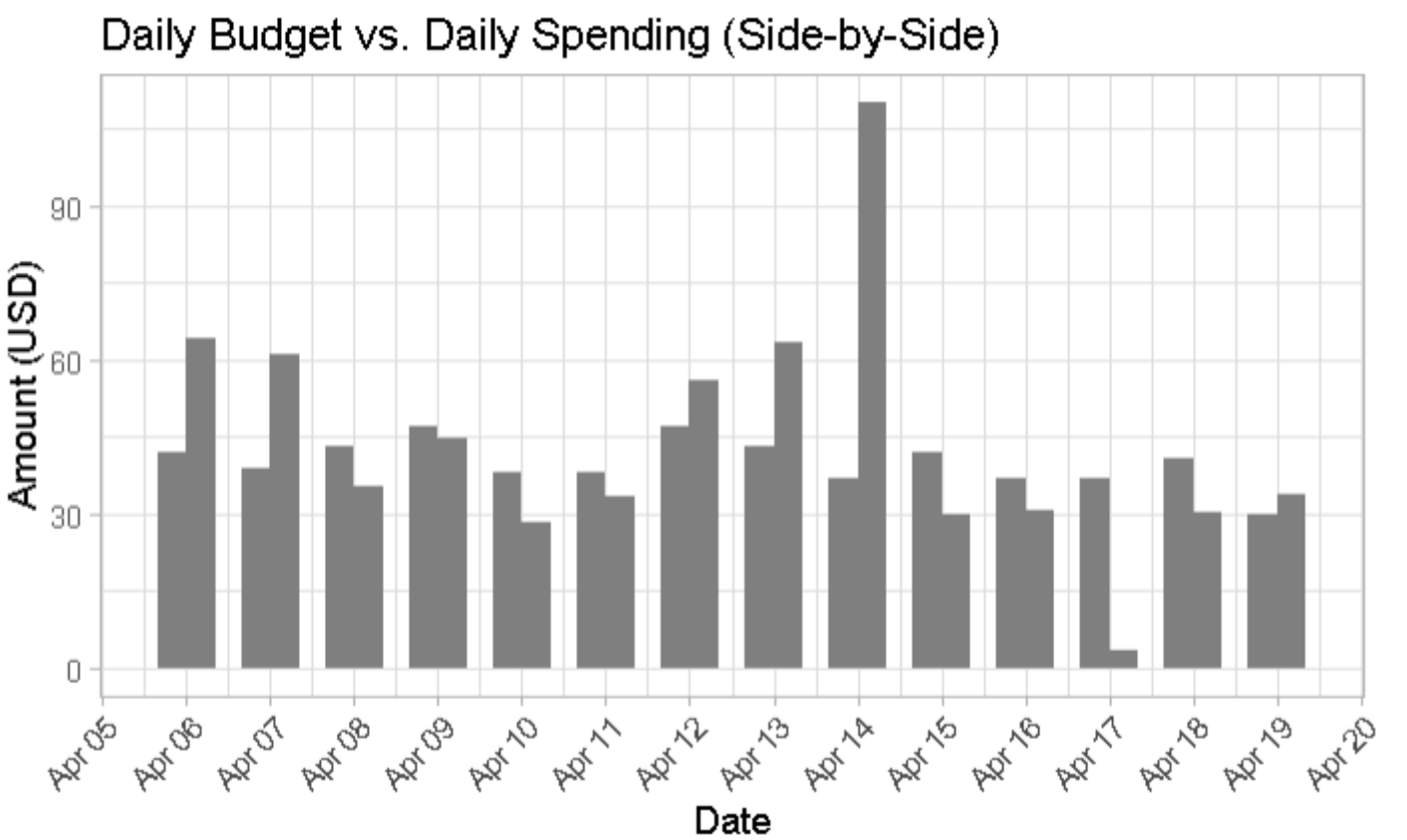
Insights

8 days under budget

6 days over budget

No days matched exactly

Largest variance: Day 14



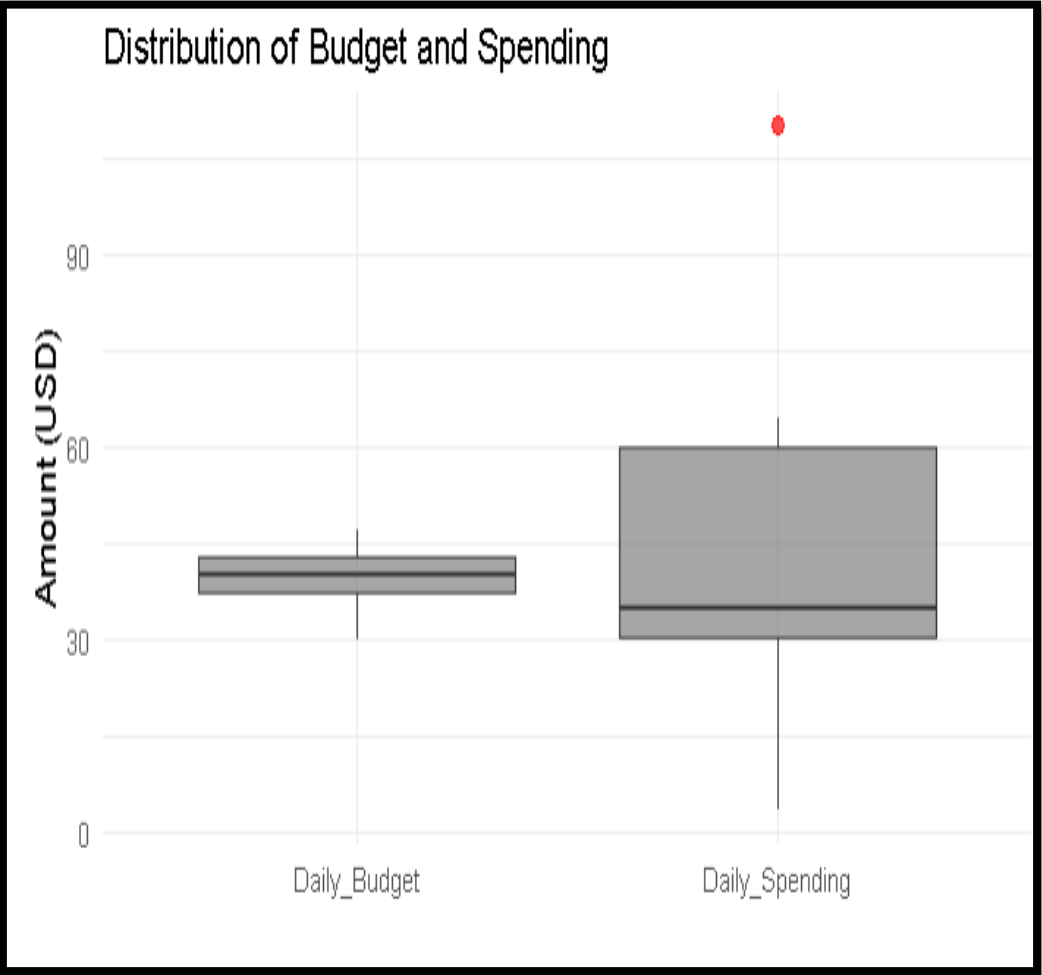
Distribution Analysis: Boxplots

Budget Distribution

Less variability in daily budget, indicating consistent financial planning.

Outlier Detection

Identification of two significant spending outliers, indicating days with exceptionally high spending.



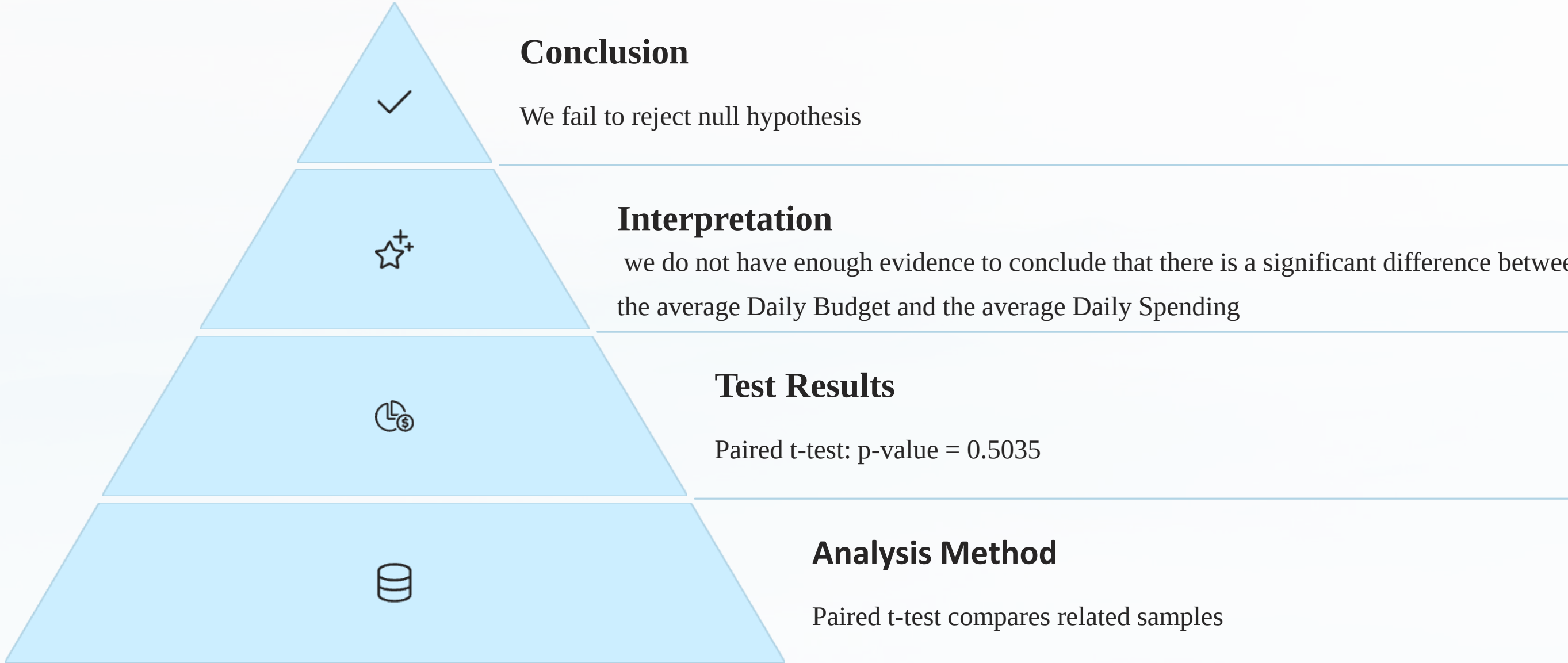
Spending Distribution

Wider range of variability in daily spending, suggesting unpredictability in spending habits.

Median Comparison

The median of daily spending appears slightly higher than the daily budget, but the difference is not statistically significant.

Statistical Analysis



Conclusion & Reflection



Key Findings

Spending showed variability, but not enough to be statistically significant.



Learning Outcomes

Statistical testing validates visual observations



Future Applications

Budget adjustments based on spending patterns